AI Adoption and Human Capital Formation in Developing Economies

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Abstract

This report develops a theoretical framework to analyze how AI Adoption with search frictions and endogenous skill acquisition in developing economies. Building on canonical task-based and search-friction models (Acemoglu, 1997; Acemoglu and Pischke, 1999; Wasmer, 2006), I introduce imperfect worker-firm matching and an ex-ante choice between portable general skills and match-specific skills. By integrating the AI-induced restructuring of tasks modelled in Acemoglu and Restrepo (2018, 2019), the model identifies three possible steady states. Notably, it exposes a *low-skill trap* where firms rely on easily accessible general AI rather than investing in complementary specific human capital. The framework generates testable predictions and policy diagnostics. A full general equilibrium closure of the model is deferred to future work.

[1]**AI Usage Acknowledgment:** Generative AI (Google Gemini) was utilized during the drafting of this report solely for LaTeX formatting assistance and copy-editing. All core economic analysis and theoretical modeling remain my (the author's) original work.

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1 Introduction

The diffusion of AI into production is reshaping the comparative advantage of human labour relative to machines. In advanced economies, a large empirical and theoretical literature documents how automation displaces workers from routine tasks while complementing workers in non-routine, cognitive roles (Autor et al., 2003; Acemoglu and Restrepo, 2019). Developing economies, however, face a structurally different challenge. Their labour markets are characterised by substantial search frictions, thin vocational training markets, and a manufacturing sector whose task content is simultaneously a target for AI substitution and a potential site for AI complementarity if, and only if, the requisite human capital is in place.

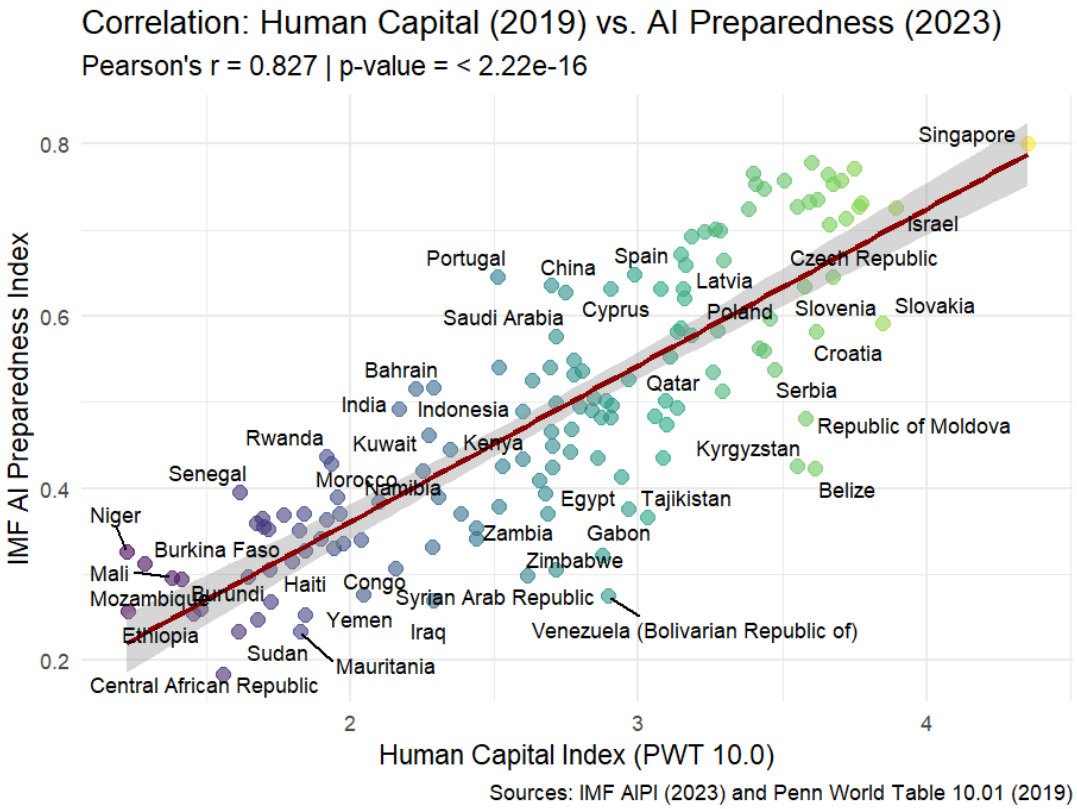


Figure 1: A cross-country comparison reveals a strong, positive correlation (Pearson's $r = 0.827$) between pre-existing human capital and current AI preparedness.

This paper is motivated by what I term as an *AI learning gap*: the possibility that the productivity gains from AI in developing economies are contingent on a form of specific human capital that neither workers nor firms have an unambiguous incentive to accumulate. If workers anticipate rapid AI-induced obsolescence of match-specific skills, they prefer investing in general (portable) skills. If firms anticipate poaching by rivals who free-ride on their training investment, they underinvest in worker development. Both dynamics can simultaneously hold, locking the economy into a low-skill trap where the

only AI in use is for general purposes, and therefore unreliable for high-value production tasks.

The canonical Acemoglu and Restrepo framework is elegant but abstracts from these frictions by assuming costless reallocation of labour across tasks (closer to a developed economy context). My addition is to relax this assumption in a tractable manner. Rather than solving a full general equilibrium model, I develop two micro-models whose mechanisms can be linked through qualitative discussion, and whose comparative statics generate empirically testable predictions.

Organisation of the paper. Section 2 reviews the literature and identifies the gap. Section 3 introduces the task taxonomy. Section 4 presents the complete Toy model, including the worker side problem (Module A) and the firm-side problem (Module B). Section 5 sketches how the modules link and characterises possible equilibria. Section 6 derives empirical predictions and policy implications. Section 7 concludes and outlines future work.

2 Literature Review

2.1 Task-Based Models of Automation

Acemoglu and Restrepo (2018) develop a task-based model in which production requires a continuum of tasks, each performed either by labour or by capital (including AI). An automation shock, modelled as expanding the set of tasks capital can perform, has two effects: a *displacement effect* that reduces labour's share of production, and a *productivity effect* that raises output and hence labour demand through scale. Acemoglu and Restrepo (2019) extend this to include the *reinstatement effect*: as automation matures, new tasks emerge that are comparative-advantage activities for labour. Together, these effects determine whether automation raises or lowers aggregate wages.

Crucially, these canonical models assume that workers can costlessly reallocate across tasks. This assumption is tenable as an approximation for short-run shocks in flexible labour markets, but it is empirically and institutionally implausible in developing economies where spatial mobility is limited, occupation-specific training is costly, and formal labour markets coexist with large informal sectors.

Aghion et al. (2017) provide a complementary framework using a CES aggregator over AI-produced and human-produced tasks. Their model clarifies that the aggregate effect of AI on labour income depends critically on the elasticity of substitution between human and AI labour—a parameter that varies across tasks and sectors and is empirically

difficult to identify.

2.2 Search Frictions and Skill Formation

Wasmer (2006) integrates the Diamond-Mortensen-Pissarides (DMP) model framework with a choice between general and specific human capital. He shows that, under search frictions, an increase in labour market tightness reduces the relative attractiveness of specific skills for workers: unemployed workers expect to find general-skill jobs quickly, diluting the option value of locking into a specific match. This result is central to my worker-side module.

Acemoglu (1997) demonstrates that wage compression under search frictions creates a positive externality from firm-sponsored training: because trained workers cannot extract the full marginal product in a frictional market, the firm retains a surplus that can recoup training costs. This provides the microfoundation for firm-sponsored training even in the absence of long-term contracts—and forms the basis of my firm-side module.

3 Task-Based Categorisation

I distinguish three types of tasks, aligned with the displacement/reinstatement logic of Acemoglu and Restrepo (2019) and explicitly grounded in the empirical exposure framework developed by the International Monetary Fund (Cazzaniga et al., 2024).

Category	Definition
HEHC (High Exposure, High Complementarity)	AI augments skilled labor; human and AI inputs are strictly complementary.
HELC (High Exposure, Low Complementarity)	Tasks automatable by general AI; human labor becomes redundant once AI reliability exceeds threshold δ.
LE (Low Exposure)	Tasks AI cannot perform, requiring manual dexterity, physical presence, or contextual judgement.

Empirical relevance: This taxonomy directly mirrors the cross-country AI exposure framework established by the IMF. As Cazzaniga et al. (2024) demonstrate in their analysis of occupational mobility, while the theoretical potential for AI augmentation

(HEHC) exists globally, its realization depends entirely on the underlying labor market structure. In advanced economies, workers displaced from HELC tasks can more easily transition into HEHC roles due to established institutional training frameworks.

In developing economies, however, the workforce is heavily skewed toward LE tasks (informal and agricultural sectors) and lacks the specific human capital required to capture the HEHC augmentation premium. Consequently, the HEHC/HELC margin becomes the critical bottleneck: it is the exact dimension along which the lack of specific skill formation traps frictional developing markets into relying on pure AI displacement rather than human-AI complementarity.

4 The Toy Model

This section presents the core model and its predictions. I develop two static decision problems, for workers and for firms, that incorporate AI adoption through the δ parameter and through modifications to the standard search-and-matching logic. .

4.1 Model Overview and Key Departures from ISM Papers

The model takes three building blocks from the papers covered in the ISM and modifies each to incorporate AI adoption. Table 1 summarises the modifications.

Table 1: Model Building Blocks and AI-Related Modifications

Source	**Original mechanism**	**AI-related modification**
Acemoglu and Restrepo (2019)	Tasks automated when AI is cheaper; workers reallocate freely	Introduce δ (AI reliability) to modulate HELC task automation; restrict reallocation via friction parameter f
Wasmer (2006)	Workers choose g or s based on wage premium and separation risk	AI raises separation risk λ^s for specific-skill matches and affects outside option through δ
Acemoglu (1997)	Wage compression enables firm training; surplus $= y^s - w^s > \tau$ (difference b.w. output and wages of worker equipped with specific skills greater than training costs to equip the worker.)	δ raises y^g (HELC output), shrinking the training surplus $y^s - y^g$; this is the AI learning gap mechanism

Note on the 'Toy" model environment: The model environment makes several simplifications to align with the scope of this report. The model is static (no Bellman equations or continuous time as in the original Wasmer paper), the equilibrium has not been solved for (wages are treated as parameters to be comparative-statically varied, not solved in closed form), and deliberately split into two parts (the worker's and the

firm's decision problems are separable). This allows each mechanism to be isolated and understood before linking them.

4.2 Modelling Production

Following Aghion et al. (2017), aggregate output is a CES composite of the three task categories:

$$Y = \left[\alpha_{\text{LE}} Y_{\text{LE}}^{\rho} + \alpha_{\text{HELC}} Y_{\text{HELC}}^{\rho} + \alpha_{\text{HEHC}} Y_{\text{HEHC}}^{\rho} \right]^{1/\rho}, \quad \rho < 1, \quad \sum_{j} \alpha_j = 1. \tag{1}$$

HELC output: This is where "general" AI (more easily available, less reliable technology) can substitute for labour. I model it as a convex combination of AI and human general-skill labour, weighted by the AI reliability index $\delta \in [0, 1]$:

$$Y_{\text{HELC}} = \delta \cdot A_{\text{AI}} + (1 - \delta) \cdot L_g, \tag{2}$$

where A_{AI} is AI output and L_g is human general-skill labour. When $\delta = 0$, only human labour can perform HELC tasks. When $\delta = 1$, AI is a perfect substitute and human labour is redundant in this category.

HEHC output: AI and specific-skill human labour are complements as neither input is productive without the other. Hence, I model this output as:

$$Y_{\text{HEHC}} = \min\{A_{\text{specific-AI}},\ \kappa L_s\}, \tag{3}$$

where L_s is specific-skill labour and $\kappa > 0$ is a productivity scale parameter. A min function specification is chosen for simplicity; any strictly complementary production function would deliver the same descriptive results this model goes on to show.

Why δ is the key policy parameter

Equation (2) captures the core tension. As general AI improves ($\delta \to 1$), HELC tasks become cheaper to automate, reducing the output premium for general-skilled human labour. But this does not automatically push workers toward HEHC tasks: HEHC tasks require specific human capital (eq. 3) that must be accumulated before production begins. The AI learning gap arises precisely when δ rises faster than specific-skill human capital can be built.

4.3 Model Setup

Workers: There is a unit mass of workers. Each worker must choose a skill type *before* searching for a job. Think of this as a one-time decision made at the start of working life. The two options are:

- **General skills** (g): Portable across all HELC tasks; freely accessible (zero cost), but the wage w^g is forced down through competition as general AI becomes a substitute for labour.
- **Specific skills** (s): Productive only in HEHC tasks [2]; incurs a one-time cost $c_s > 0$ (think of this as the time and money spent learning to work with a particular AI platform), and carries a risk λ^s of the match becoming obsolete if the firm's specific AI stack/platform of choice changes.

Firms: There is a large number of identical firms [3]. Each firm posts one vacancy and, upon matching with a worker, decides whether to *train* the worker for HEHC tasks (specific skills, cost τ) or use the worker in HELC tasks (no training). Once trained, the worker can be poached by a rival firm with probability $\phi \in [0, 1]$; the training cost is sunk.

Labour market friction: Workers and firms meet imperfectly. The probability that an unemployed worker finds a job in any period is $f \in (0, 1)$, which I treat as an exogenous matching rate. Think of f as summarising labour market tightness. a higher f means more vacancies relative to workers, making it easier for workers to find jobs. The poaching probability ϕ on the firm's part is increasing in f.

Notation summary:

$\delta \in [0, 1]$	AI reliability (higher = better general AI)
$c_s > 0$	Worker's cost of acquiring specific skills
$\tau > 0$	Firm's cost of training a worker
$\lambda^g \geq 0$	Rate at which general-skill matches become obsolete
$\lambda^s \geq 0$	Rate at which specific-skill matches become obsolete
$\phi \in [0, 1]$	Probability that a trained worker is poached
$f \in (0, 1)$	Worker's probability of finding a job
w^g, w^s	Wages in general-skill and specific-skill jobs
$y^g(\delta), y^s$	Output per worker in HELC (AI-augmented) and HEHC tasks
r	Worker's discount rate over full time horizon

[2] may not be a realistic assumption as skills needed to operate a specific AI platform/stack may still be portable

[3] this model assumes a representative firm, heterogeneity is reserved for future work

4.4 Module A: The Worker's Skill Choice

4.4.1 The Worker's Decision Problem

A worker must compare the expected lifetime payoff from choosing general skills versus specific skills. The key insight from Wasmer (2006) is that job search is imperfect. A worker who chooses specific skills and does not match with a training firm earns nothing from those skills. This creates a risk of *mismatch* (risk of investing in specific skills that are never used).

Expected payoff from general skills. If a worker chooses g, they find a HELC job with probability f and earn wage w^g for as long as the job lasts. With probability $(1-f)$ they remain unemployed and receive flow value b (home production or benefits). Discounting future payoffs at rate r and assuming an expected job duration of $1/\lambda^g$ (general jobs become obselete at rate λ^g), the discounted expected payoff is:

$$\begin{aligned} V^g &= \int_0^\infty be^{-rt}dt + f \cdot \int_0^\infty (w^g - b)e^{-(r+\lambda^g)t}dt \\ V^g &= \frac{b}{r} + f \cdot \left(\frac{w^g - b}{r+\lambda^g}\right) \end{aligned} \tag{4}$$

giving the equation:

$$V^g = f \cdot \frac{w^g - b}{r+\lambda^g} + b/r. \tag{5}$$

Expected payoff from specific skills. If a worker chooses s, they find a training firm with probability f (the same matching rate), but must also pay the skill cost c_s upfront. If matched, they earn $w^s > w^g$ but face a higher obsolescence rate $\lambda^s \geq \lambda^g$ (AI may change the firm's technology, making the worker's specific knowledge redundant faster):

$$\begin{aligned} V^s &= \int_0^\infty be^{-rt}dt + f \cdot \int_0^\infty (w^s - b)e^{-(r+\lambda^s)t}dt - c_s \\ V^s &= \frac{b}{r} + f \cdot \left(\frac{w^s - b}{r+\lambda^s}\right) - c_s \end{aligned} \tag{6}$$

giving the equation:

$$V^s = f \cdot \frac{w^s - b}{r+\lambda^s} + b/r - c_s. \tag{7}$$

4.4.2 The Skill-Choice Condition

The worker chooses specific skills if and only if $V^s \geq V^g$, i.e.:

$$\underbrace{f \cdot \frac{w^s - b}{r + \lambda^s}}_{\text{discounted surplus from } s} - c_s \geq \underbrace{f \cdot \frac{w^g - b}{r + \lambda^g}}_{\text{discounted surplus from } g}. \tag{8}$$

Cancelling f (it appears on both sides) and rearranging:

$$\boxed{\frac{w^s - b}{r + \lambda^s} - \frac{w^g - b}{r + \lambda^g} \geq \frac{c_s}{f}.} \tag{9}$$

Interpreting this condition:

Left-hand side: the wage-surplus premium of specific skills over general skills, discounted by a function of the respective obsolescence rates. This is the *benefit* of acquiring specific skills.

Right-hand side: the training cost c_s per unit of job-finding probability. When the labour market is tight (f is high), this cost is effectively lower. Workers expect to use their specific skills frequently, so the upfront cost is well-amortised. When the market is not tight (f is low), the cost per effective use of training is high.

4.4.3 How AI Adoption (through δ) Enters the Worker's Decision

The key insight is that δ affects w^g directly, through the HELC production function (2). As δ rises (general AI improves), HELC output per worker increases, and Nash bargaining (as followed in the Wasmer paper), the wage w^g rises. This narrows the left-hand side of (9). Formally:

Proposition 1 (AI Substitution Reduces Specific-Skill Investment)**.** *Suppose wages are set by Nash bargaining with worker bargaining power β, so that $w^j = \beta y^j + (1-\beta)b$. Then $w^g(\delta) = \beta[\delta A_{\text{AI}} + (1-\delta)L_g] + (1-\beta)b$ is increasing in δ. An increase in AI reliability δ therefore reduces the left-hand side of (9), making specific-skill investment less attractive for workers, ceteris paribus.*

Interpretation: When general AI gets better, it pushes up wages for general-skill workers (because AI makes their tasks more productive). This reduced the comparative attractiveness of the riskier, costlier specific-skill path. Workers rationally shift toward g, but this is exactly the skill type that AI is about to replace.

Additionally, a shock to AI obsolescence directly raises λ^s (the rate at which specific-skill

matches break down) because as firms' AI platforms evolve, workers' learned complementary skills become redundant more quickly. This compresses the discounted surplus (by increasing $\frac{w^s-b}{r+\lambda^s}$) on the left-hand side of (9), further discouraging specific-skill investment.

4.5 Module B: The Firm's Training Decision

4.5.1 The Firm's Decision Problem

When a worker is matched to a firm, this model assumes that the firm observes whether the worker decided to invest in general or specific skills before searching for a job. [4] Then, a matched firm chooses between two strategies:

(1) **No training**: Assign the worker to HELC tasks. The firm earns output $y^g(\delta) = \delta A_{\text{AI}} + (1-\delta)L_g$ per period (AI does part of the work), pays wage w^g to worker, and retains the worker until separation at rate λ^g.

(2) **Training**: Invest τ upfront to equip the worker with specific skills. The worker can then perform HEHC tasks, generating output $y^s > y^g$ per period. The firm pays wage $w^s > w^g$, but retains the worker only until either obsolescence (rate λ^s) or poaching (probability ϕ).

4.5.2 Firm's Expected Profits

No training (HELC):

$$\Pi^g = \int_0^\infty (y^g(\delta) - w^g)e^{-(r+\lambda^g)t}dt$$
$$\Pi^g = \frac{y^g(\delta) - w^g}{r+\lambda^g} \tag{10}$$

giving the equation:

$$\Pi^g = \frac{y^g(\delta) - w^g}{r+\lambda^g}. \tag{11}$$

This is output minus wage, discounted by a function of the worker's discount rate and job tenure ($1/(r+\lambda^g)$).

[4] This assumption follows from the model on search frictions and wage compression in Acemoglu (1997). The assumption may be unrealistic, but has been made for tractability. Acemoglu and Pischke modify it in a different model in Acemoglu and Pischke (1999), which focuses on asymmetric information as the mechanism behind wage compression.

Training (HEHC):

$$\Pi^s = -\tau + (1-\phi) \cdot \int_0^\infty (y^s - w^s) e^{-(r+\lambda^s)t} dt + \phi \cdot (0)$$
$$\Pi^s = -\tau + (1-\phi) \cdot \frac{y^s - w^s}{r + \lambda^s} \tag{12}$$

giving the equation:

$$\Pi^s = -\tau + (1-\phi) \cdot \frac{y^s - w^s}{r + \lambda^s}. \tag{13}$$

The firm pays τ upfront, but with probability ϕ the worker is poached before the firm recoups this cost. With probability $(1-\phi)$, the firm earns the discounted surplus from a specific-skill match.

4.5.3 The Training Condition

The firm trains if and only if $\Pi^s \geq \Pi^g$:

$$\boxed{(1-\phi) \cdot \frac{y^s - w^s}{r + \lambda^s} - \tau \quad \geq \quad \frac{y^g(\delta) - w^g}{r + \lambda^g}.} \tag{14}$$

Interpreting this Condition:

Left-hand side: Can be interpreted as the expected net return to training, which is the discounted surplus from a specific-skill match, adjusted for poaching risk, minus the firm's training cost.

Right-hand side: Can be interpreted as the opportunity cost of training, which is the surplus the firm would earn from simply assigning the worker to HELC tasks (which are increasingly productive as AI improves).

Training is profitable when the HEHC surplus *net of poaching and training costs* exceeds the HELC surplus. This is the Acemoglu (1997) wage-compression mechanism: search frictions keep w^s below y^s, so the firm captures a surplus that makes training worthwhile.

4.5.4 How AI Adoption (through δ) Enters the Firm's Decision

Now it is clear how δ creates the AI learning gap:

Proposition 2 (AI Reliability Erodes the Training Incentive)**.** *Since $y^g(\delta)$ is increasing in δ (eq. 2), an increase in AI reliability raises the right-hand side of (14), the opportunity cost of training. The training condition becomes harder to satisfy, and for δ sufficiently large, no firm trains.*

Threshold: Training is viable only if $\delta < \delta^*$, where δ^* is implicitly defined by:

$$(1-\phi)\cdot\frac{y^s-w^s}{r+\lambda^s}-\tau=\frac{y^g(\delta^*)-w^g}{r+\lambda^g}. \tag{15}$$

For $\delta > \delta^*$, HELC production is cheap enough that no firm invests in training, and the economy is stuck in the low-skill trap.

4.6 Comparative Statics: Both Modules

Table 2 collects the comparative statics for both modules.

Table 2: Comparative Statics of the Toy Model

Parameter change	**Effect on worker: chooses s?**	**Effect on firm: trains?**	**Mechanism**
$\delta\uparrow$ (better general AI)	Less likely (LHS of (9) shrinks via $w^g\uparrow$)	Less likely (RHS of (14) grows via $y^g\uparrow$)	AI learning gap
$\lambda^s\uparrow$ (faster AI obsolescence)	Less likely (discounted s-surplus falls)	Less likely (expected tenure falls)	Obsolescence risk
$\tau\uparrow$ (training cost rises)	Less likely (Skilling cost for the worker (c_s) increases)	Less likely (direct effect)	Cost effect
$\phi\uparrow$ (more poaching)	Ambiguous	Less likely ($(1-\phi)$ falls)	Poaching related problem
$f\uparrow$ (tighter market)	More likely (RHS of (9) falls via c_s/f)	Less likely ($\phi\uparrow$ with f)	"Goldilocks" range
$y^s\uparrow$ (HEHC productivity)	More likely ($w^s\uparrow$)	More likely (LHS grows)	Complementarity premium

The most important feature of Table 2 is the last column for f: tighter labour markets help workers but hurt firms (via poaching), creating a *Goldilocks range* of labour market tightness in which both conditions (workers choosing specific-skilling and firms choosing to train) hold simultaneously.

5 Solving the Model

5.1 Linking the Two Modules

The worker and firm problems are explicitly linked because:

(i) Wages w^g and w^s appear in both conditions; they depend on the equilibrium pool of skills (if many workers choose s, competition among firms for specific-skill workers drives up w^s, affecting both conditions).

(ii) Labour market tightness f appears in both conditions and is determined by how many vacancies (for jobs involving general-skill tasks vs specific-skill tasks) firms post, which in turn depends on expected profits from training vs. not training.

For this toy model, I do not solve for these endogenous wages and tightness in closed form. Instead, I treat (w^g, w^s, f) as parameters and use the two conditions to describe plausible equilibrium configurations.

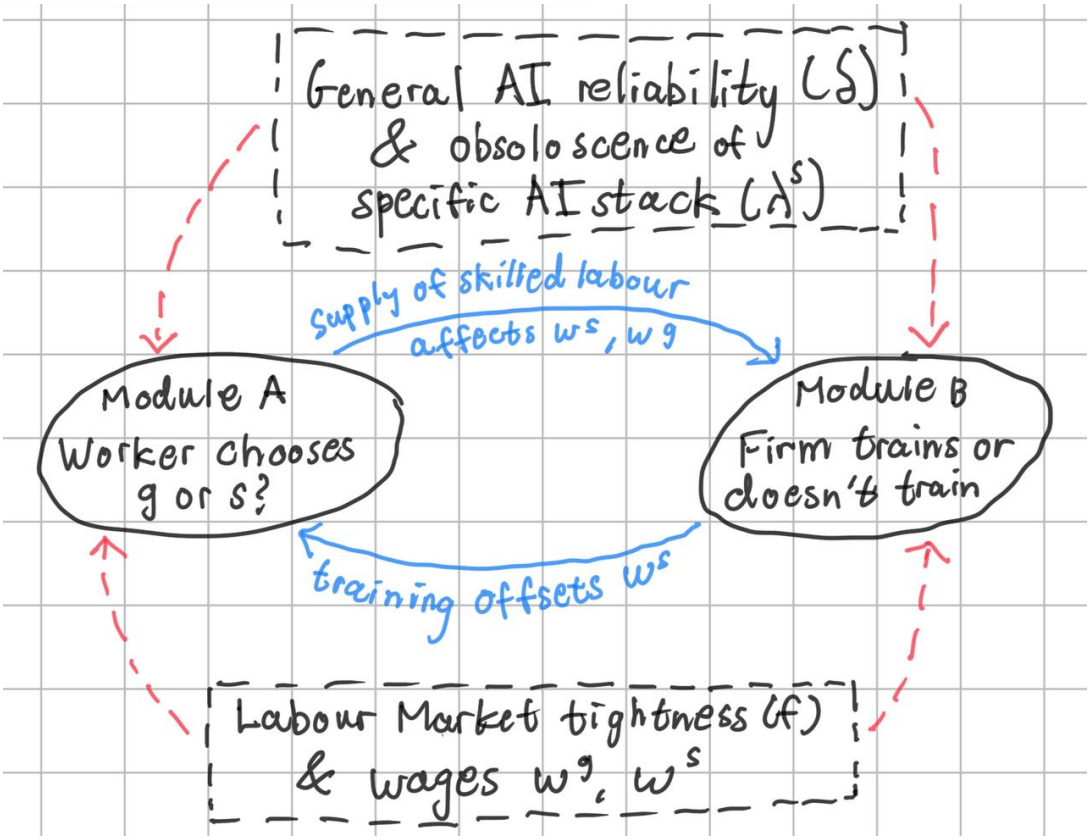


Figure 2: Interdependencies in the toy model. Solid arrows represent direct interactions between the worker and firm problems. Dashed arrows represent how the common parameters (δ, λ^s, f, w^g, w^s) enter both conditions simultaneously.

5.2 Three Plausible Equilibria

Equilibrium 1: High-Skill (Desirable) Both conditions hold: workers choose s, firms train. This requires: $\delta < \delta^*$ (general AI not yet a strong substitute), λ^s moderate, f in the Goldilocks range, and $y^s - y^g$ large enough to cover τ. This equilibrium is self-reinforcing: workers invest in specific skills because they expect training offers; firms train because they face a pool of specific-skill candidates.

Equilibrium 2: Low-Skill Trap Neither condition holds: workers choose g, firms do not train. This requires $\delta > \delta^*$ or extreme f (too tight, so poaching deters training; too slack, so c_s/f is prohibitively high). Self-reinforcing in the same way: workers do not invest because no training is offered; firms do not train because no one has invested in specific skills.

Equilibrium 3: Dual Economy Some firms train (those with low τ or high y^s) and some do not. A fraction of workers invest in specific skills; the rest choose g. This produces a bifurcated economy with a productive specific-skill sector and a HELC sector competing directly with AI. This is likely the empirically most relevant configuration for developing economies.

Proposition 3 (Multiplicity)**.** *For parameters in a range where both the high-skill equilibrium and the low-skill trap satisfy their respective conditions, the economy admits at least two steady states. Which equilibrium is reached depends on expectations: if workers expect training offers they invest in s; this makes training profitable for firms, which confirms their expectation.*

5.3 Graphical Analysis of Equilibria

The graphs provided below are heuristic sketches; they are not drawn to scale using calibrated numerical values, but rather serve to visually illustrate the comparative statics and the resulting macroeconomic phase space.

Figure 3 illustrates the worker's decision problem, plotting the expected lifetime payoff (V) against labor market tightness (f).

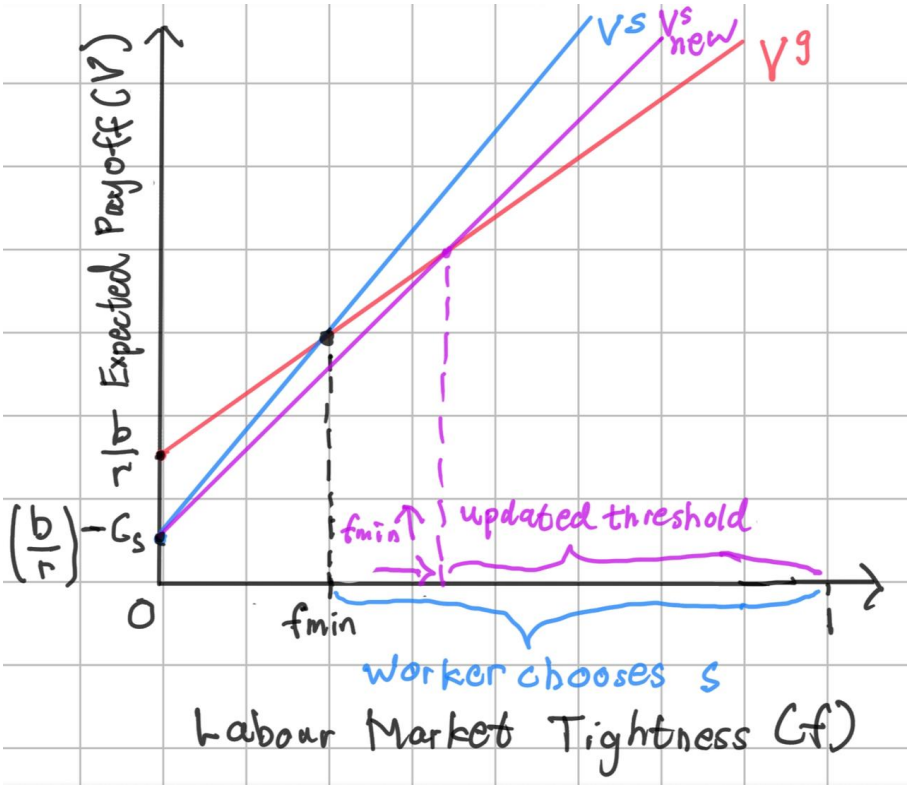


Figure 3: The Worker's Skill Choice

The baseline value of acquiring general, portable skills is represented by the upward-sloping line V^g, which originates at b/r. The expected payoff for acquiring specific skills, V^s, originates at a lower intercept, $(b/r) - c_s$, reflecting the upfront cost of learning specialized skills. Because specific skills command a higher wage premium ($w^s > w^g$), the V^s curve is steeper.

The worker rationally chooses specific skills only when $f > f_{min}$. The comparative statics demonstrate the effect of an AI obsolescence shock: if the specific obsolescence rate (λ^s) increases, the V^s curve flattens to V^s_{new}. This shifts the threshold to the right, illustrating that workers demand a tighter labor market to compensate for the heightened risk of acquiring non-portable skills.

Figure 4 illustrates the firm's training decision, plotting expected profit (Π) against general AI reliability (δ).

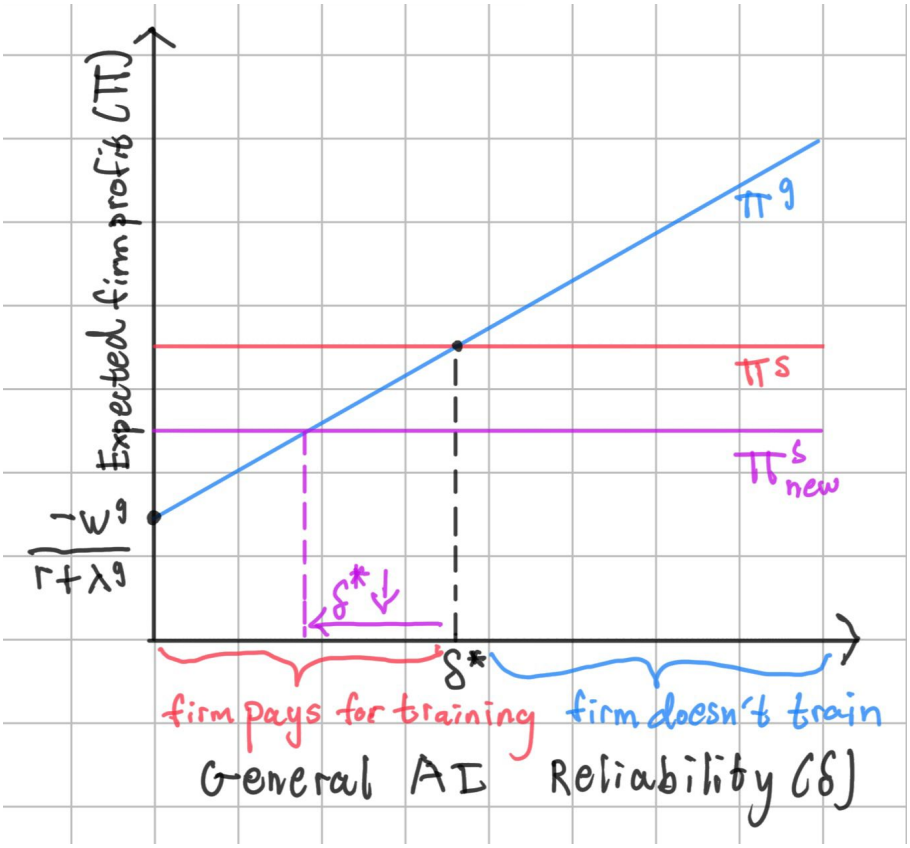


Figure 4: The Firm's Training Decision

The expected profit from training a worker for HEHC tasks (Π^s) is horizontal, as specific human-AI complementarity does not depend on the reliability of general AI. Conversely, the expected profit without training (Π^g) slopes upward; as general AI becomes more reliable, the firm extracts greater surplus from untrained workers performing HELC tasks.

Firms are only willing to pay the sunk cost of training (τ) if $\delta < \delta^*$. The comparative statics highlight the vulnerability of this investment: if the training cost τ increases, the Π^s line shifts downward to Π^s_{new}, pulling the threshold leftward to $\delta^{*\downarrow}$. This visually captures the "AI Learning Gap," where marginally better outside options (cheaper general AI) crowd out firm-sponsored training.

Figure 5 synthesizes the constraints from both modules into a unified diagram, mapping regions where equilibria occur.

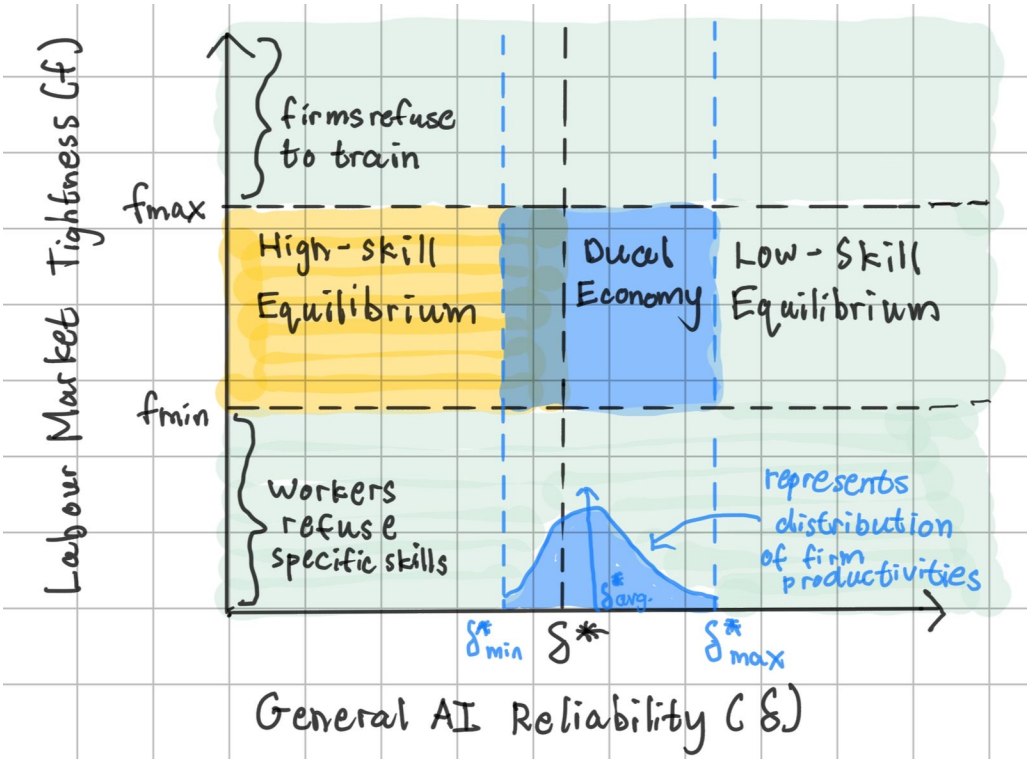


Figure 5: Phase Diagram of equilibria

The graph maps the economy across the parameters of labor market tightness (Y-axis) and general AI reliability (X-axis):

- **High-Skill Equilibrium (Reinstatement):** The yellow region represents the High-Skill Equilibrium. The labor market is tight enough for workers to pay c_s ($f > f_{min}$), but slack enough that firms are not deterred by poaching ($f < f_{max}$). Simultaneously, general AI is not reliable enough to displace specific training.
- **Frictional Traps:** The regions below f_{min} and above f_{max} represent states where labor market frictions unilaterally destroy the incentive for either the worker or the firm to invest in HEHC tasks, regardless of AI capability.
- **The Dual Economy:** This presents a possibility if the model were to incorporate firm heterogeneity, depicted by the distribution of firm productivities on the X-axis. Consequently, the firm training threshold is not a single point but a band between δ^*_{min} and δ^*_{max}. In the blue zone, AI reliability has crossed the threshold for less productive firms (pushing them into a Low-Skill Equilibrium), while highly productive firms continue to invest in specific training.
- **Low-Skill Equilibrium (Displacement):** As δ surpasses δ^*_{max}, even highly productive firms abandon training, and the entire economy relies on general AI without human capital formation to perform HEHC tasks.

6 Policy Implications

6.1 Testable Predictions

The model generates four testable predictions for future empirical work:

P1. Sectors experiencing faster improvement in general AI (higher δ) should show declining firm-sponsored training rates, ceteris paribus.

P2. Countries with intermediate labour market tightness (moderate f) should show higher specific-skill investment and AI-complementary human capital than countries at the extremes.

P3. The AI learning gap should be widest in lower-middle-income countries, where the augmentation potential is high but specific-skill human capital is lowest.

6.2 Policy Levers

Training subsidies (reduce τ): Reducing the firm's training cost directly relaxes condition (14). However, training subsidies are effective *only* when f is in the Goldilocks range: if poaching risk ϕ is too high (very tight market), subsidies simply transfer rents to poaching firms.

AI reliability standards: Regulatory mandates requiring human oversight for high-stakes decisions effectively cap δ, preventing it from crossing δ^*. By keeping HELC output below HEHC output, such mandates preserve the productivity gap that makes training profitable.

Anti-poaching measures (reduce ϕ): Non-solicitation clauses or sector-level training levies reduce ϕ, raising the effective return on training. The trade-off is reduced labour market fluidity; the policy is most justified in sectors where training costs τ are high and HEHC productivity premiums $y^s - y^g$ are large.

Industrial policy targeting HEHC sectors: Subsidies for AI-complementary production (e.g., advanced manufacturing segments that use AI for quality control with human supervisors) raises y^s and shifts the training condition toward feasibility. This is distinct from simply subsidising AI adoption, which raises δ and can have the opposite effect.

7 Conclusion and Future Work

This report developed a tractable toy model to analyse the interaction between AI-induced task restructuring, search frictions, and skill formation in developing economies. The central contribution is the identification of the *AI learning gap*: a parameter range in which improvements in general AI reliability (δ) simultaneously erode both workers' incentive to acquire specific skills and firms' incentive to train, potentially trapping the economy in a low-skill, high-general-AI equilibrium.

Future work:

(i) Close the model with endogenous wages (Nash bargaining) and endogenous tightness (free entry of vacancies) to derive formal existence of equilibria and multiplicity proofs.

(ii) Calibrate to developing-country data (e.g., Indian or Indonesian firm surveys on AI adoption and training expenditure) to quantify the size of the learning gap.

(iii) Extend to heterogeneous firms (drawn from a distribution of τ or y^s) to formally model the dual-economy equilibrium.

(iv) Trace the transition path as δ rises over time, asking whether there exist policies that can prevent a good equilibrium from tipping into the low-skill trap.

8 Appendix: Nash Bargaining and Free Entry

For closing the model, this theoretical summary of key concepts used to endogenise wages and tightness may be used:

Nash bargaining wages (As given in Wasmer (2006)): With worker bargaining power $\beta \in (0, 1)$:

$$w^j = \beta\, y^j + (1 - \beta)\, b, \qquad j \in \{g, s\}. \tag{16}$$

Substituting into conditions (9) and (14) pins down wages as functions of primitives.

Free entry (As given in Wasmer (2006)): Firms post vacancies at cost k per vacancy. The free-entry condition equates expected profit to vacancy cost:

$$f \cdot \max\{\Pi^g,\, \Pi^s\} = k, \tag{17}$$

where f is determined by the matching function. This equation, combined with the two decision conditions, closes the model and pins down f endogenously.

Steady-state employment. With separation rates λ^g, λ^s and matching rate f, steady-state employment shares satisfy:

$$f \cdot u = \lambda^g e^g + (\lambda^s + \phi)\, e^s, \qquad u + e^g + e^s = 1, \tag{18}$$

where u, e^g, e^s are unemployment, general-employment, and specific-employment shares, respectively.

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